

Richard Tairouz

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LinkedIn | GitHub | Portfolio

Education

Florida International University

B.S. Applied Mathematics & B.S. Computer Science

Miami, FL

Expected December 2026

- GPA: **3.95/4.00**
- Florida International University Honors College
- Florida Bright Futures Scholarship recipient
- Relevant coursework: Mathematical Modeling, Probability and Statistics, Machine Learning, Data Structures and Algorithms, Operating Systems, Linear Algebra, Differential Equations, and Numerical/Computational Methods

Research Experience

Florida State University

Research Experience for Undergraduates (REU)

Tallahassee, FL

May 2025 – May 2026

- Conducted mathematical and computational research involving stochastic modeling and quantitative analysis.
- Developed computational methods to study stochastic systems and investigate model behavior under varying assumptions.
- Implemented numerical experiments and analyzed model outputs using scientific programming techniques.
- Presented research findings at a research symposium and contributed to a written research paper.

FIU Discovery Lab

Undergraduate Researcher

Miami, FL

Jan 2023 – Jun 2023

- Investigated computational and mathematical concepts related to quantum computing and quantum algorithms.
- Studied the mathematical foundations of quantum computation, including linear algebra and probabilistic measurement.
- Explored computational approaches for solving problems using quantum-computing frameworks.

Selected Research & Technical Projects

Power Series Stochastic Modeling

Python

- Developed computational models for stochastic processes using mathematical series representations and numerical simulation.
- Analyzed model behavior computationally and compared numerical results across different parameter configurations.

Guitar Chord Recognition

Python, Machine Learning

- Developed a machine-learning pipeline to classify 12 major chords, 12 minor chords, and noise from audio-derived features.
- Evaluated multiple classification algorithms, including Random Forest, K-Nearest Neighbors, and Support Vector Machines.

3D Face Modeling & Analysis

Python, Computer Vision

- Developed computational techniques for processing and analyzing three-dimensional facial data.
- Applied mathematical and algorithmic methods to extract and evaluate geometric characteristics from 3D models.

Teaching Experience

Florida International University

Learning Assistant – Precalculus

Miami, FL

Through December 2025

- Supported students in developing mathematical reasoning and problem-solving skills in precalculus.

- Explained mathematical concepts through multiple approaches and assisted students during collaborative problem-solving sessions.

Technical Skills

Programming: Python, R, C, C++, C#, Java, SQL, JavaScript, TypeScript, HTML/CSS

Scientific & Data: NumPy, Pandas, SciPy, scikit-learn, Matplotlib, statistical modeling, machine learning, numerical methods, data analysis

Software & Tools: Git, GitHub, Docker, Linux, MySQL, React, $\text{\LaTeX}$, VS Code

Mathematics: Probability, stochastic modeling, linear algebra, differential equations, mathematical modeling, numerical computation, statistics

Research Interests

Stochastic Modeling • Machine Learning • Scientific Computing • Applied Mathematics • Optimization • Computational Modeling